occurs in the reactor vessel, so no external steam generator is required, and that the fuel elements are enclosed in fuel boxes to improve the flow pattern. Although the cores are about the same size for PWRs and BWRs, the reactor vessel is much taller for the BWR; typical values are 22 m in height and 6 m in diameter. Since the pressure in the vessel is lower for BWRs than for PWRs, the vessel wall can be thinner (a typical value is 16 cm). BWRs usually contain more uranium, but with lower enrichment, than PWRs. While only about one-fifth of the core is replaced each year in a BWR, about one-third to one-fourth of the core is replaced annually for a PWR; in both cases this corresponds to about 30 t of used fuel replaced every year.

The BWR is controlled by blades entering from the bottom of the reactor vessel. The blades are usually of cruciform cross section and fit in between the fuel boxes surrounding the fuel elements. The fuel elements often contain burnable poison (e.g. Gd_2O_3). Because of the boiling in the tank boric acid cannot be used for reactivity control; however, it is provided as an extra emergency shutdown feature. The control blades contain boron carbide rods or powder placed in channels in the stainless steel blade-wings. In addition to negative temperature coefficients both for fuel and moderator, BWRs have a negative *void coefficient*. This means that bubble formation (*void*) along the fuel pins reduces the reactivity. As mentioned in §19.6 it is therefore common practice to control the reactor power by the main coolant circulation pumps; increased circulation initially reduces the voids (and fuel temperature) thereby increasing power production until the previous void fraction has been almost restored at a slightly higher average fuel temperature. However, a very large negative void coefficient should also be avoided as it can cause an undesirable amplification of pressure fluctuations in the reactor vessel.

19.14. POWER STATION EFFICIENCY

Not all of the heat produced in a nuclear reactor can be used for work, i.e. for turning the turbine blades connected to the rotor of an electric generator. According to the second law of thermodynamics, the maximum *thermal efficiency* (η) is

$$\eta = (Q_{in} - Q_{out})/Q_{in} \tag{19.42}$$

where Q_{in} is the heat input into a machine and Q_{out} is the heat discharged from that machine. Most reactors operate according to the *Rankine cycle* in which a liquid medium is heated and vaporized at constant pressure, and work is carried out through adiabatic expansion. While the steam entering the turbine should be as dry as possible to reduce turbine blade wear, the steam on the backside is wet due to some condensation. This effect is normally reduced by dividing the turbine into a high pressure and a low pressure part with intermediate reheating by primary steam. Because steam is not an ideal gas (an accurate calculation of Q from heats of vaporization and the ideal gas law is impossible) empirically based steam tables must be used.

The highest thermodynamic efficiency is achieved in the *Carnot cycle* in which energy input (heating the working medium) and work both occur at different but constant temperatures, T_{in} and T_{out}. For a "Carnot engine"

$$\eta_{max} = (T_{in} - T_{out})/T_{in} \tag{19.43}$$

A typical PWR plant may deliver steam of ~275°C to the turbine and have an outlet temperature from the condenser on the turbine backside of ~29°C. Using (19.43) $\eta_{max} = [(275 + 273) - (29 + 273)]/(275 + 273) = 0.45$. A typical BWR usually delivers steam at ~285°C and has a condenser temperature of ~25°C, hence $\eta_{max} = 0.47$. Due to a less efficient energy cycle, friction, heat losses, pumps, etc., the net efficiency (η_{net}) of both reactor types is only about 0.32−0.35 (net electric output delivered to the grid divided by gross thermal output from reactor). In coal-, oil- and gas-fired power plants higher steam temperatures can be achieved, ~500°C with T_{in} 530°C and T_{out} 30°C, $\eta_{max} = 0.65$ leading to an η_{net} ~0.50.

If we compare η_{net} for the two kinds of power plants we find that the fossil fuel power plant has about 40% higher efficiency, i.e. the amount at "*reject heat*" is up to 40% higher for the typical nuclear power plant ((0.50−0.30)/0.50). This lower efficiency is a consequence of the lower steam temperature of the nuclear reactor. However, in the last 10 years considerable development has taken place with respect to coolant, cladding and fuel stability, and inertness, allowing considerably higher fuel temperatures, cf. Ch. 21.

The thermal efficiency of any reactor can be raised by using some of the heat directly for a beneficial purpose, such as for processing in the chemical industry or for district heating. If the diverted heat is at a higher temperature than normal for the condenser coolant effluent such diversion will reduce electricity output correspondingly. If this "lost" electricity had been used to drive heat pumps instead, the total energy efficiency had only increased by a small amount, or not at all, by the direct diversion of heat. In Russia some smaller nuclear power stations produce district heat and electricity for local consumption; EGP reactors in Table 19.1.

The performance characteristics of power stations are commonly described by the availability and load factors, and by the forced outage factor. The *availability factor* is the time the station has been available for operation divided by the length of the desired time period. Thus if it has been desired to operate the station for 6000 h, but it has only been possible to run it for 5500 h (because of repair, etc.) the availability factor is 92%. Typical availability factors are 95−100% for water power, 75−85% for conventional fossil power, and ~70% for nuclear power (usually the reactor has a higher availability, but with a lower value of ~70% for the turbine). Availability factors >80% have been achieved for LWRs.

The *load factor* is the ratio between the electrical net energy delivered by the station and the net electrical energy it would have produced running at maximum power (i.e. design power multiplied by the time). A 1000 MW_e power station which has produced

5.80 TWh during the year has had a load factor of 66% (= 5.80×10^6 MWh per 1000 MW $\times$ 8760 h y^{-1}). When nuclear power is used for *base load* (i.e. running at full power when operating), its load factor is only limited by how well the nuclear power stations perform. When nuclear power is not used for base load, the load factor depends not so much on the power station performance but rather on the local energy demand, available alternative energy sources and their price. Thus in Sweden during 1999 the annual load factor of hydro power was ~50%, thermal power ~22%, wind power ~19%, and nuclear power ~80%. The low load factors for hydro power and thermal power are due to the use of these in "follow the load" mode, whereas nuclear power was used for base load and wind power operated at its momentary capacity. The rather low value for thermal power is caused by its cost of operation, which is higher than the other alternatives; wind power is subsidized to make it competitive. Typical world wide average yearly load factors for all power reactors weighted with production capacity are (2011 figures): PWR 81%, BWR 73%, PHWR 71%, and RBMK 79%. These figures do not include reactors of less than 150 MWe or data from utilities with less than 4 operating reactors.

A better picture of the power station performance in situations where the demand may be less than the installed capacity is given by the *forced outage rate* which is the ratio between the number of hours a power station has been shut down because of malfunction and the total number of hours during that period. A power station which has been shut off for one month in January for repair of the main feed water pump and three months (June, July, and August) because of no demand for electricity, during which time part or the fuel elements were changed, has an 8% forced outage rate (i.e. one month out of twelve). The forced outage rate is very similar for all power stations of similar size and age. Values for new 1000 MW_e BWR or PWR stations vary from 15% up; a typical average value is 25%, normally decreasing with time.

19.15. REACTOR SAFETY

Greater than 95% of all the operation shutdowns in nuclear power stations are due to failures common to conventional thermal power stations. However, in this text we consider only safety with regard to the nuclear steam supply system which account for the other 5%. We can distinguish between three safety (or risk) levels: (i) small deficiencies due to imperfect technique, which are inevitable in any human enterprise (e.g. small "chronic" releases of radioactivity to the environment), (ii) accidents, which normally should not occur with good equipment, but still are "probable" (e.g. pipe breaks, fires, etc.) and for which protective measures must be included in the reactor design (design basis accident, DBA), and (iii) maximum credible accident (MCA) which is the most serious reactor accident that can reasonably be imagined from any adverse combination of equipment malfunction, operating errors, and other foreseeable causes.